

Problem #2: Abracadabra (15 marks)

Solution:

Assignment 1:

The given sentence has the following 7 decompositions.

1. *ab-ba-rad-rac-rad-ab-ab-adar-ba* [the always blue of blue the the teacher always]
2. *ab-ba-rad-rac-radab-ab-adar-ba* [the always blue of respect the teacher always]
3. *ab-ba-rad-racrad-ab-ab-adar-ba* [the always blue grow the the teacher always]
4. *ab-bar-adrac-rad-ab-ab-adar-ba* [the good student blue the the teacher always]
5. *ab-bar-adrac-radab-ab-adar-ba* [the good student respect the teacher always]
6. *ab-baradra-cra-dabab-adar-ba* [the morning sun enchant(s) teacher always]
7. *ab-baradra-crada-babad-arba* [the morning sky glow(s) red]

Assignment 2:

Only the decompositions 5, 6 and 7 are meaningful. Possible English translations are:

5: *ab-bar-adrac-radab-ab-adar-ba* = The good student always respects the teacher.

6: *ab-baradra-cra-dabab-adar-ba* = The morning sun always enchants teachers OR
The morning sun always enchants the teacher.

7: *ab-baradra-crada-babad-arba* = The morning sky glows red.

Assignment 3:

ab-baradra-crada-babad-arba

Assignment 4:

Yes, it is possible. For example, the Abracadabra sentence – *radabra* has a valid decomposition: *rad-abra*, but the program would decompose it as *radab-ra*, which is invalid.
